

AIGIS Retail Bond Planning Tool

AIGIS零售債券規劃工具

Kenny Chan

Danny Chan

Open Data Hackathon 2025 Smart Economy

Main Beneficiaries

主要受惠人



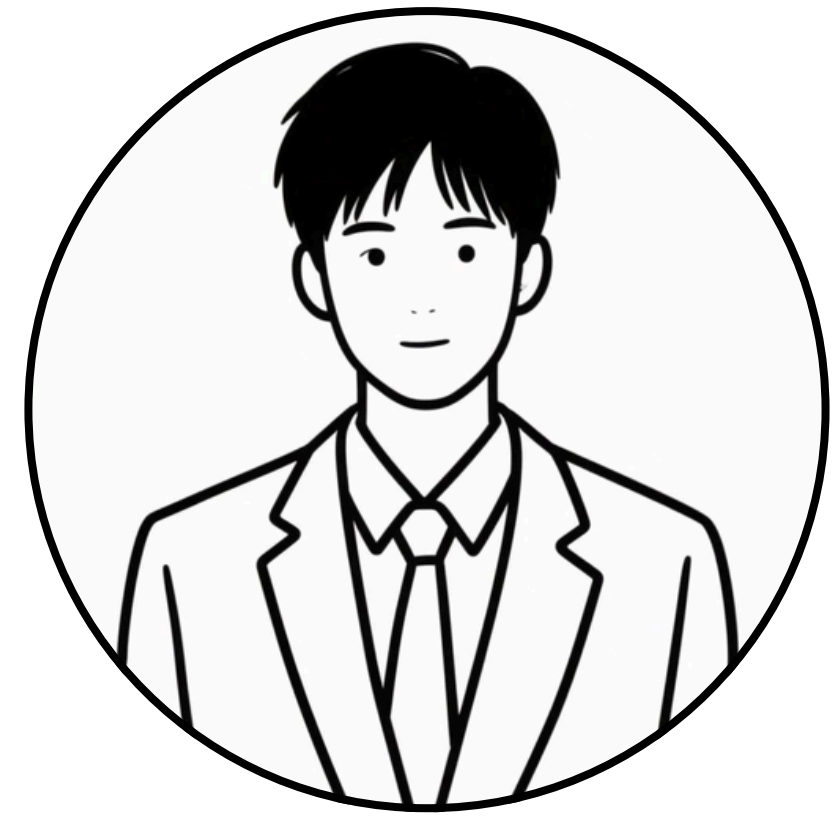
Interest earners

食息一族



Investment banks

投資銀行



Bond issuers

發債人

Public Pain Point in Hong Kong

香港民生痛點

庫房冇錢，

咪發債Cap水。

The government has no money, so it issues bonds to raise funds.



Elderly
阿公

有錢都拎曬去出糧，

都冇野幫到市民。

Even with money, most of it goes to salaries — nothing left for the public.



Working Class
基層

淨系識Cap水

我都識做啦。

Just issuing bonds?
Anyone can do that.

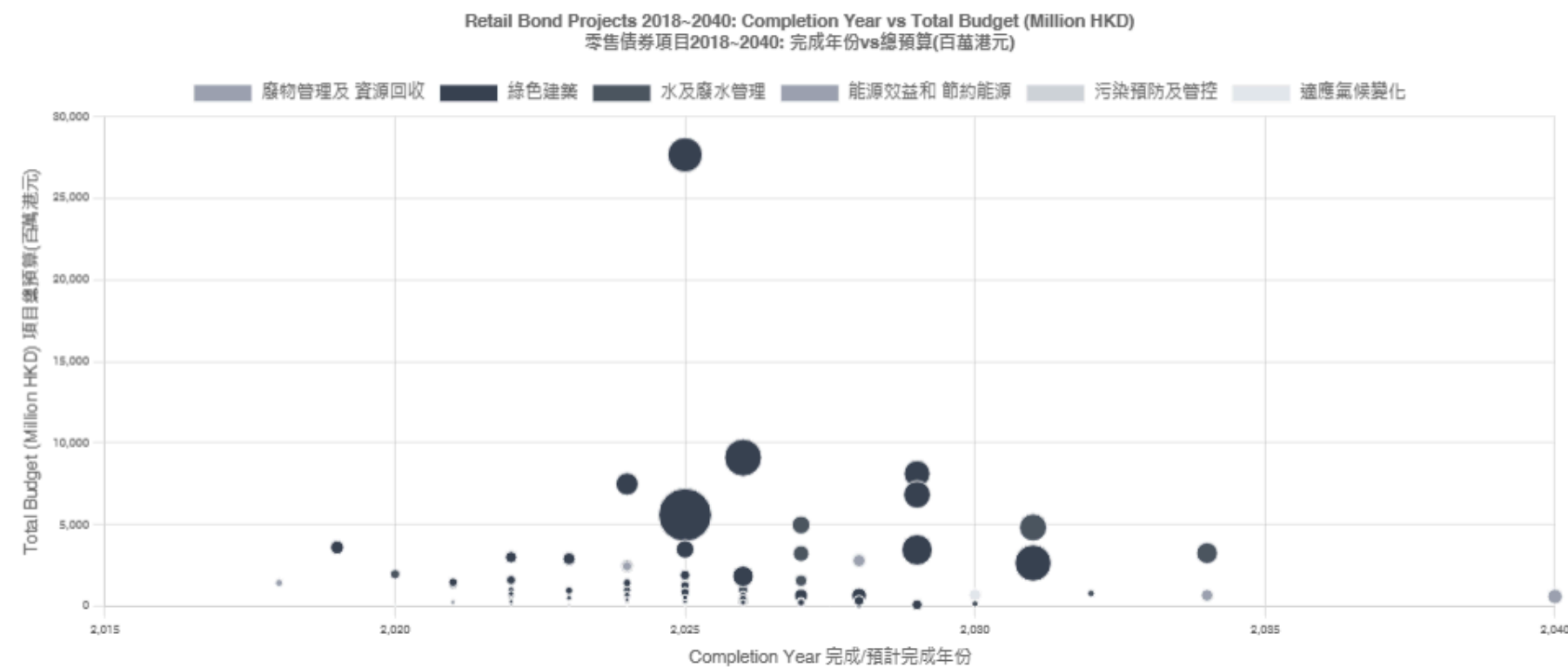
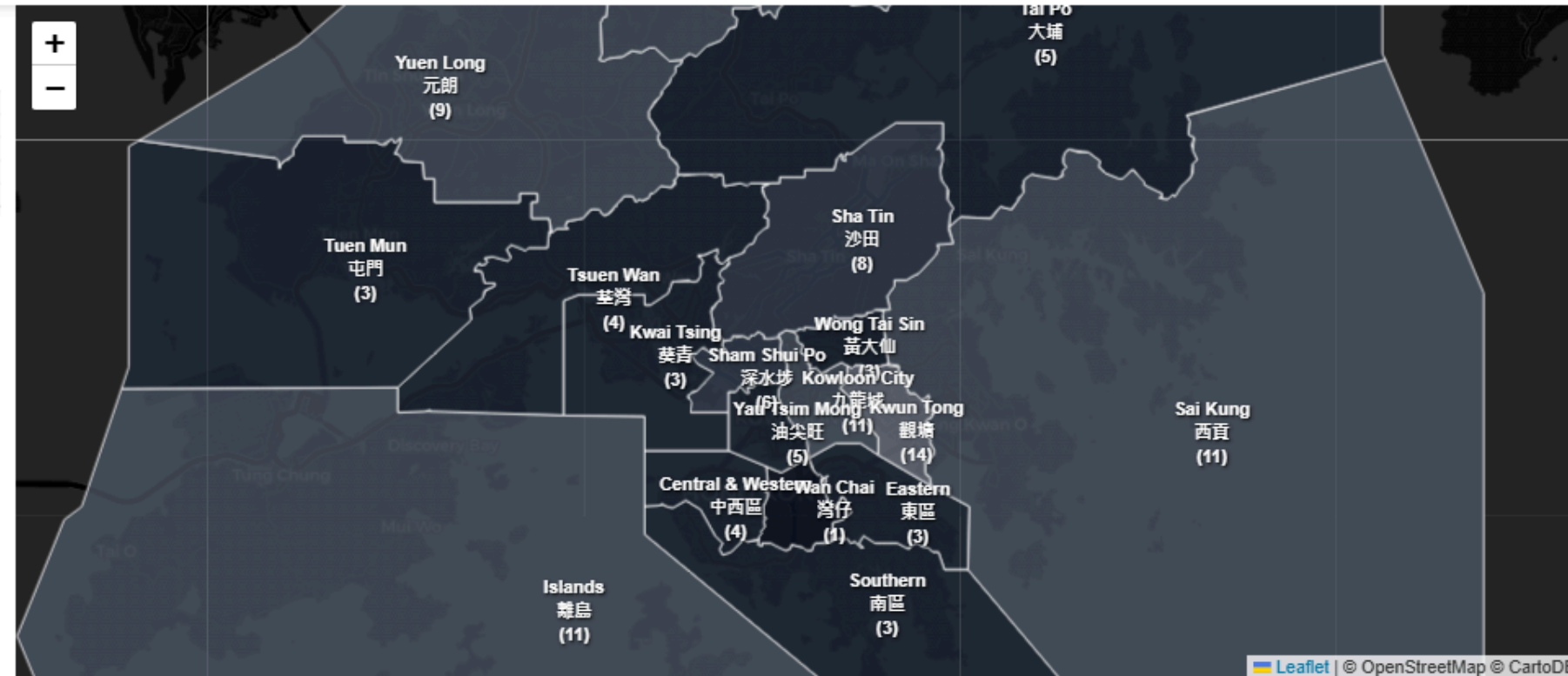


Middle Class
中產

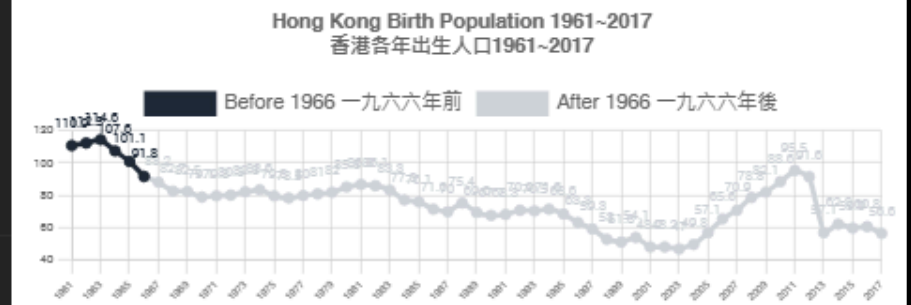
AIGIS Retail Bond Planning Tool

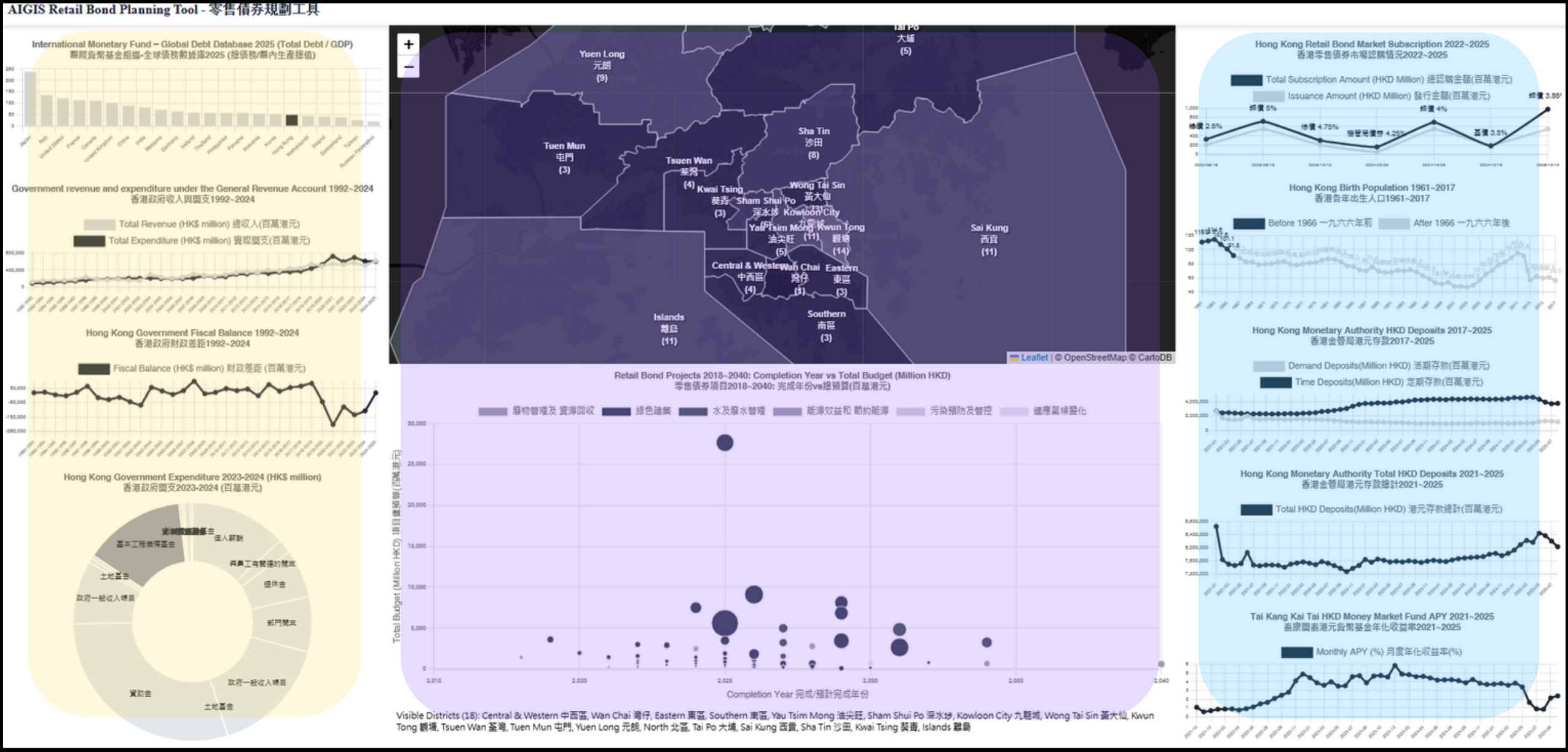
AIGIS零售債券規劃工具

AIGIS Retail Bond Planning Tool - 零售債券規劃工具



Visible Districts (18): Central & Western 中西區, Wan Chai 灣仔, Eastern 東區, Southern 南區, Yau Tsim Mong 油尖旺, Sham Shui Po 深水埗, Kowloon City 九龍城, Wong Tai Sin 黃大仙, Kwun Tong 觀塘, Tsuen Wan 荃灣, Tuen Mun 屯門, Yuen Long 元朗, North 北區, Tai Po 大埔, Sai Kung 西貢, Sha Tin 沙田, Kwai Tsing 葵青, Islands 離島



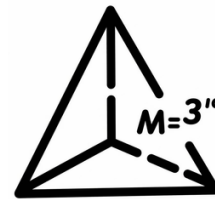


香港債務儀

債務 vs 增長
與民生幸福關係

Hong Kong Debt Dashboard

Debt vs Growth
Connected to Public Well-being

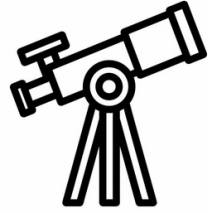


香港財政債務GIS地圖

大型基建 vs 小型民生
看得見錢花在哪

GIS Map of Hong Kong's Fiscal Debt

Big Infrastructure vs Small Community Projects
Clear Where the Money Is Spent



零售債券需求預測模型

承銷時機 & 定價規模
哪類項目最吸引市民

Retail Bond Demand Forecast Model

Timing & Pricing for Bond Sales
Projects That Attract the Most Public Interest

Hong Kong Public & Private Companies Data

香港公開數據 & 私企數據

[國際貨幣基金組織] 全球債務數據庫 Global Debt Database

https://www.imf.org/external/datamapper/GG_DEBT_GDP@GDD/SWE

[香港金融管理局] 綠色債券報告2025

<https://www.hkgb.gov.hk/tc/greenbond/greenbondreport.html>

[財經事務及庫務局/立法會發展事務委員會] 基本工程儲備基金 按總目列出的支出

<https://www.budget.gov.hk/2024/chi/pdf/c-cwrf-s.pdf>

[香港金融管理局] 金融數據月報 銀行體系 客戶存款 港元及外幣存款

<https://data.gov.hk/tc-data/dataset/hk-hkma-t03-t030301customer-deposits-by-type-hkd-fc>

<https://apidocs.hkma.gov.hk/chi/documentation/market-data-and-statistics/monthly-statistical-bulletin/banking/customer-deposits-by-type-hkd-fc/>

[香港交易所/盈透證券] 泰康開泰港元貨幣基金 年化收益率

<https://ifp.hkex.com.hk/fund/BRG187>

[前沿資本] 零售債券 市場認購

https://github.com/kenny-chen/open_data_hackathon_2025_smart_economy_001/blob/web/app/%E5%89%8D%E6%B2%BF%E8%B3%87%E6%9C%AC-%E9%9B%B6%E5%94%AE%E5%82%B5%E5%88%B8-%E5%B8%82%E5%A0%B4%E8%AA%8D%E8%B3%BC.csv

[政府統計處] 政府的收入和開支

<https://data.gov.hk/tc-data/dataset/hk-censtatd-tablechart-340-47001>

[政府統計處] 政府開支 一般收入帳目及各基金

https://www.censtatd.gov.hk/tc/web_table.html?id=340-47012

[香港金融管理局] 固定收益及貨幣市場發展路線圖2025

<https://www.hkma.gov.hk/media/chi/doc/key-information/press-release/2025/20250925c3a1.pdf>

[民政事務總署] 地方行政區分界

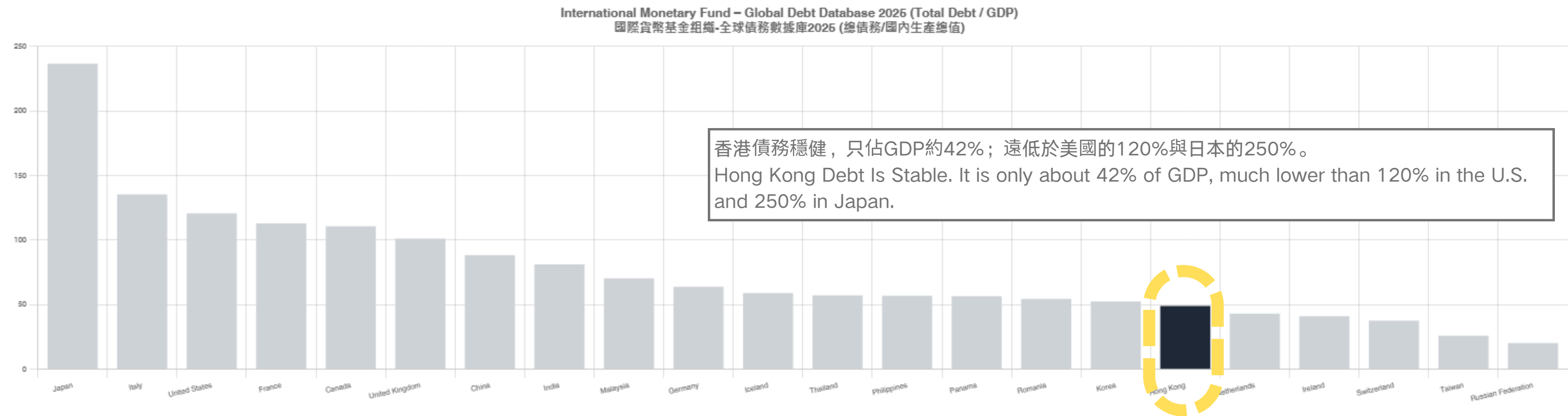
<https://data.gov.hk/tc-data/dataset/hk-had-json1-hong-kong-administrative-boundaries>



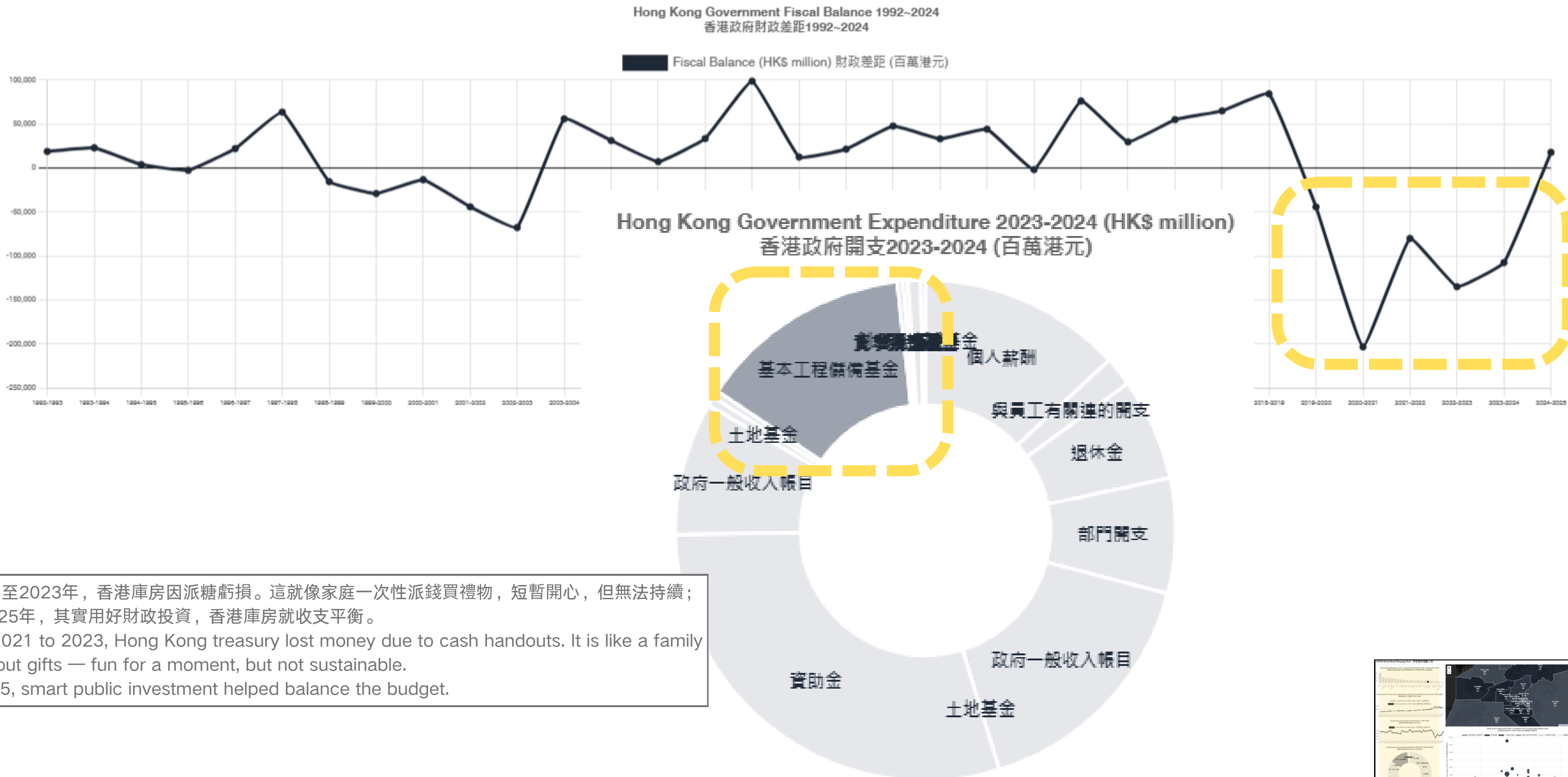
Hong Kong Debt Dashboard

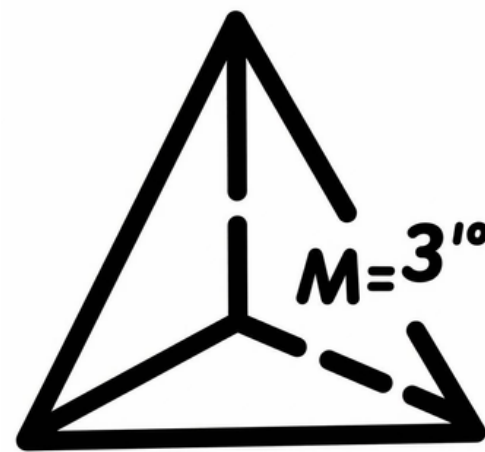
香港債務儀

Debt is a by-product of economic growth – All developed countries have debt. 債務是經濟增長的副產品。因所有發達國家都有債務。



Spending on public projects brings lasting happiness. 將錢用於財政投資，幸福感才會長久。





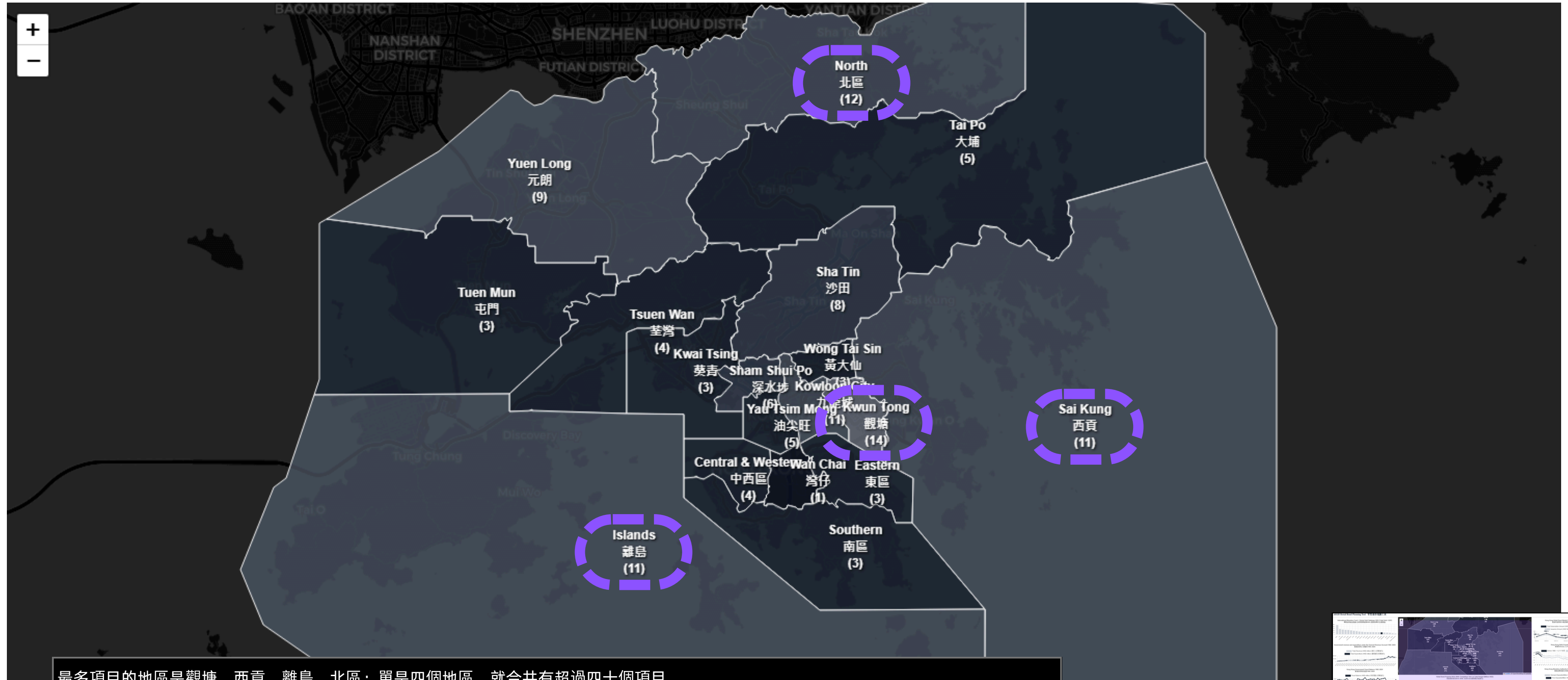
GIS Map of Hong Kong's Fiscal Debt

香港財政債務GIS地圖

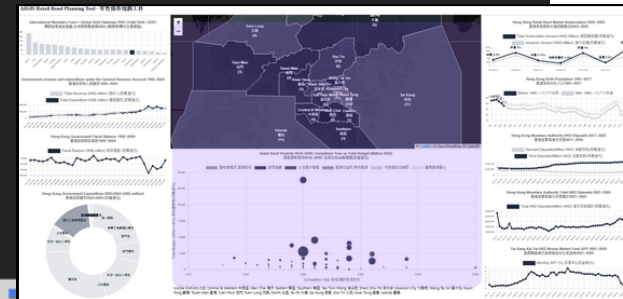
Retail bond money was used for projects across Hong Kong.

But why can't people see or feel the benefits?

全港有用到零售債券錢興建項目。但為何市民看不見、感受不到幸福？



最多項目的地區是觀塘、西貢、離島、北區；單是四個地區，就合共有超過四十個項目。
Kwun Tong, Sai Kung, Islands District, and Northern District have the most projects—over 40 in total across just these four areas.



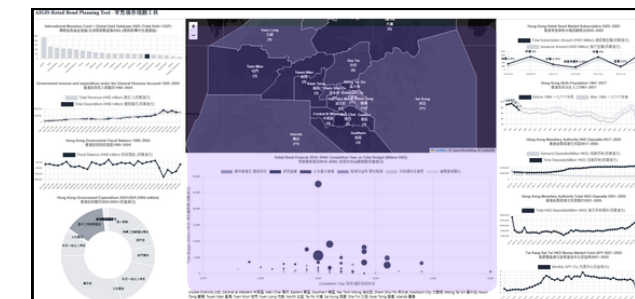
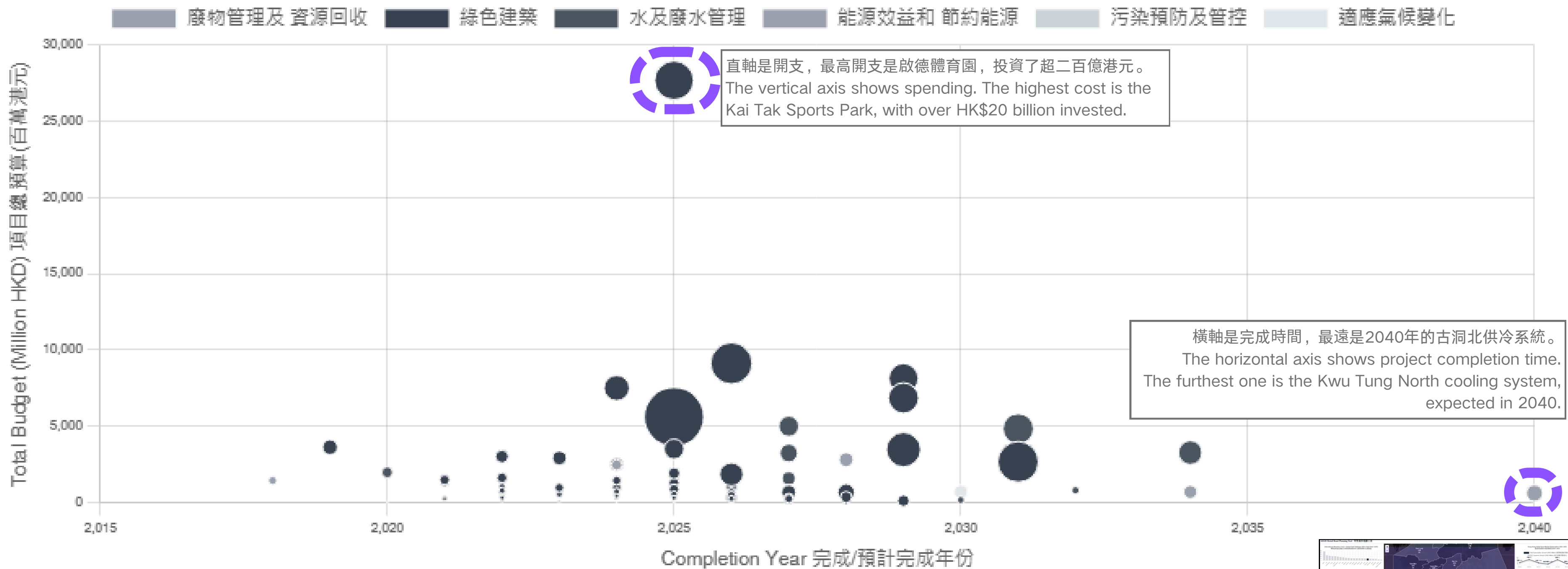
Small community projects make up less than 15%.

That's why people feel the money is spent on things that are “too big and too far away.”

小型民生僅佔不足十五%。市民自然覺得錢花得「太高太遠」。

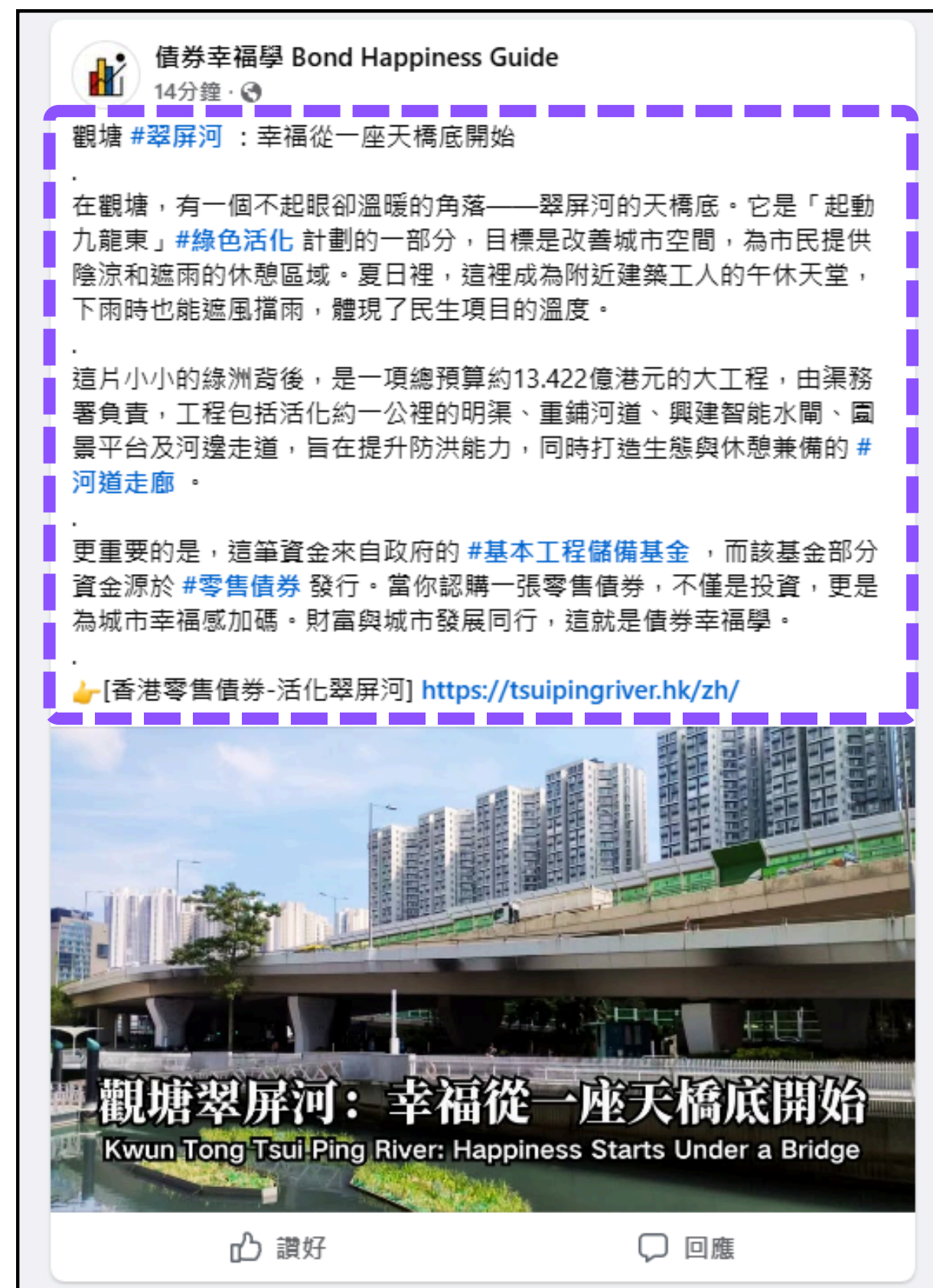
Retail Bond Projects 2018~2040: Completion Year vs Total Budget (Million HKD)

零售債券項目2018~2040: 完成年份vs總預算(百萬港元)



The happiness of bonds. Retail bonds fund civil engineering to make city life better.

債券幸福學。零售債券撥款到土木工程，提升城市生活幸福感。



#生活幸福感
#綠色活化
#河道走廊
#基本工程儲備基金
#零售債券

#Sense-of-WellBeing
#GreenRevitalization
#RiverCorridor
#CapitalWorksReserveFund
#RetailBonds

Visible results. Retail bonds drive urban development.

Under the bridge at Tsui Ping River in Kwun Tong.

看得見成果，零售債券倡導城市發展好。觀塘翠屏河的天橋底。

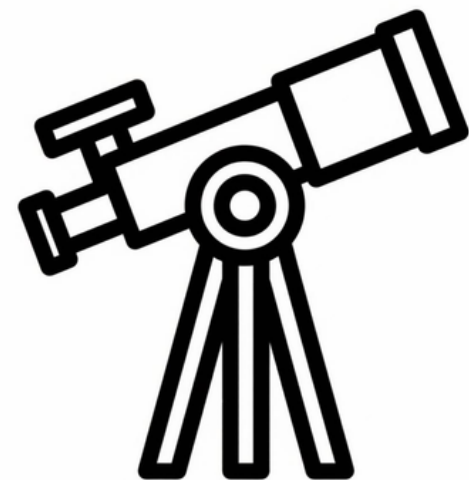


觀塘翠屏河的天橋底，是起動九龍東的綠色活化項目。雖然這只是天橋底，但這是夏天陰涼位，下雨有頂蓋，讓一群附近工作的建築工人在這午休的地方。

Tsui Ping River Flyover in Kwun Tong. It is part of the green revitalization of Kowloon East.

Even though it is just under a bridge, it is cool in summer and covered when it rains — making it a popular lunch spot for nearby construction workers.



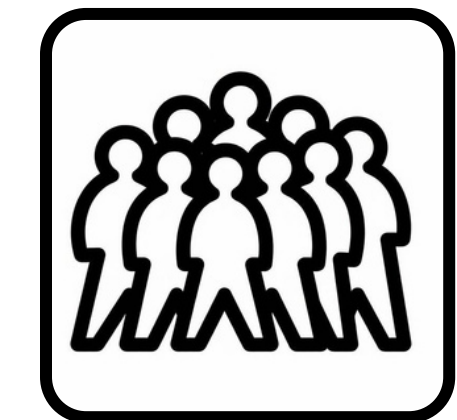


Retail Bond Demand Forecast Model

零售債券需求預測模型

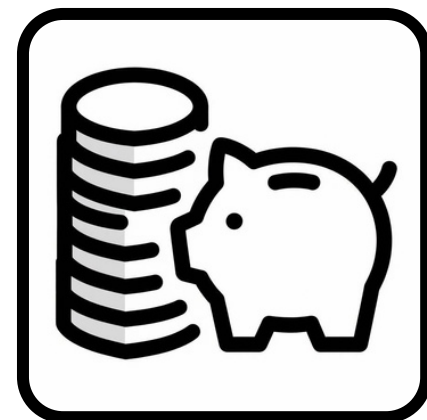
Retail Bond Demand Forecast Model

零售債券需求預測模型



人口基數
Population

×



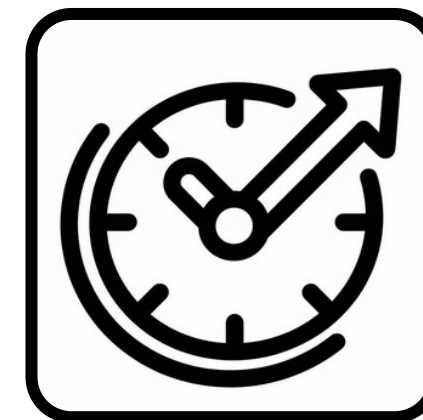
家庭存款
Household Savings

+



定價水位
Pricing Level

+



承銷時機
Timing

+



投標項目
Project Type

銀行M2存款
上升

Bank M2 Deposits Are
Rising

現金寶年化
收益率+半釐

Annual return of
cash fund + 0.5%

股市高水&
股市低水

When stock market is
high or low

大型基建+
小型民生

Big infrastructure + Small
community projects

24基債 未足額認購0.8倍

24 Infrastructure: Undersubscribed at 0.8 ×

25銀債 超額認購1.8倍

25 Silver: Oversubscribed by 1.8 ×

2026 Retail Bond Forecast

預測 2026年零售債券

 超額1.8倍 1,000億港元

Oversubscription 2.1 × for HK\$100 billion

 超額1.1倍 600億港元

Oversubscription 1.2 × for HK\$60 billion

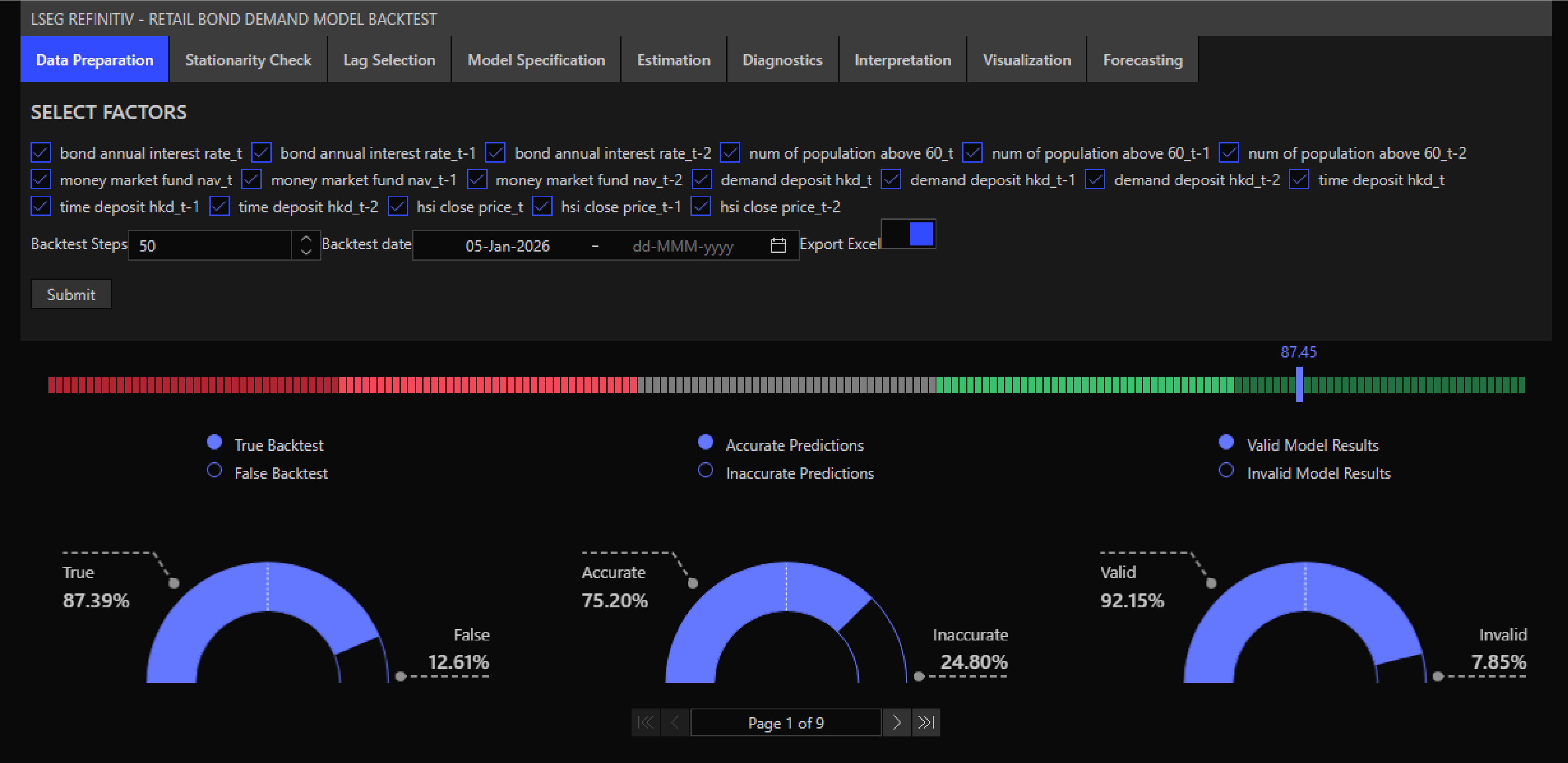
	食息一族 Interest Earners	家庭存款 Household Savings	定價水位 Pricing Level	承銷時機 Timing	投標項目 Project Type
 上限建簿 Upper Bookbuilding	上升88.2萬人 Up to 882,000 people	上升2.2yoy Up 2.2% YoY	3.5%	28,000 points Hang Seng Index	不變 No change
 下限建簿 Lower Bookbuilding		不變0.0yoy Flat at 0.0% YoY	2.8%	24,000 points Hang Seng Index	

基礎財富指標
Basic Wealth Indicator

吸引力指標
Attractiveness Indicator

Backtesting accuracy is over 82%, using the LSEG Refinitiv financial terminal.

回測準確度超82%+, 利用LSEG Refinitiv路孚特金融終端機。



87.45

● True Backtest

○ False Backtest

● Accurate Predictions

○ Inaccurate Predictions

● Valid Model Results

○ Invalid Model Results

True

87.39%

False

12.61%

Accurate

75.20%

Inaccurate

24.80%

Valid

92.15%

Invalid

7.85%

Page 1 of 9

Find the financial and economic data that most affect retail bond subscriptions,
build a backtesting model, and show the results visually.

找出「零售債券認購額」最大影響力的金融經濟數據， 建立回測模型並可視覺化結果。

Data Loading &
Preprocessing
資料載入與前處理

Stationarity
Testing
平穩性檢定

Lag Selection
滯後階數選擇

Distributed Lag
Model (DLM)
分佈滯後模型

Results &
Visualization
結果與視覺化

forward fill 前向填補

$$x_t = \begin{cases} x_t & \text{if } x_t \text{ is known} \\ x_{t-1} & \text{if } x_t \text{ is missing} \end{cases}$$

Augmented Dickey – Fuller

ADF單根檢定

$$\Delta y_t = \alpha + \beta t + \gamma y_{t-1} + \sum_{i=1}^p \delta_i \Delta y_{t-i} + \varepsilon_t$$

Autoregressive Model

自回歸

$$y_t = c + \sum_{i=1}^p \phi_i y_{t-i} + \varepsilon_t$$

Distributed Lag Model

分佈滯後模型

$$y_t = \alpha + \sum_{j=0}^k \beta_j x_{t-j} + \varepsilon_t$$

Lag Coefficient Distribution

Bar Chart

滯後係數分佈條形圖

$$\beta_{mj} \text{ from the model: } y_t = \alpha + \sum_{m=1}^M \sum_{j=0}^k \beta_{mj} x_{m,t-j} + \varepsilon_t$$

log transform 對數轉換

$$x'_t = \log(x_t)$$

Kwiatkowski Phillips

Schmidt Shin

KPSS檢定

$$KPSS = \frac{\sum_{t=1}^T S_t^2}{T^2 \hat{\sigma}^2}$$

Akaike Information Criterion

赤池信息准則

$$AIC = 2k - 2 \ln(\hat{L})$$

Bayesian Information Criterion

貝氏資訊準則

$$BIC = \ln(n) \cdot k - 2 \ln(\hat{L})$$

P-value

P值

$$p = P(T \geq |t_{\text{obs}}| \mid H_0)$$

T-test statistic

T檢定統計量

$$t = \frac{\hat{\beta} - \beta_0}{SE(\hat{\beta})}$$

Heatmap of Coefficients

係數熱圖

$$\text{Coefficient Matrix } C = [\beta_{mj}]_{M \times (k+1)}$$

Cumulative Impact Ranking

累積影響力排名

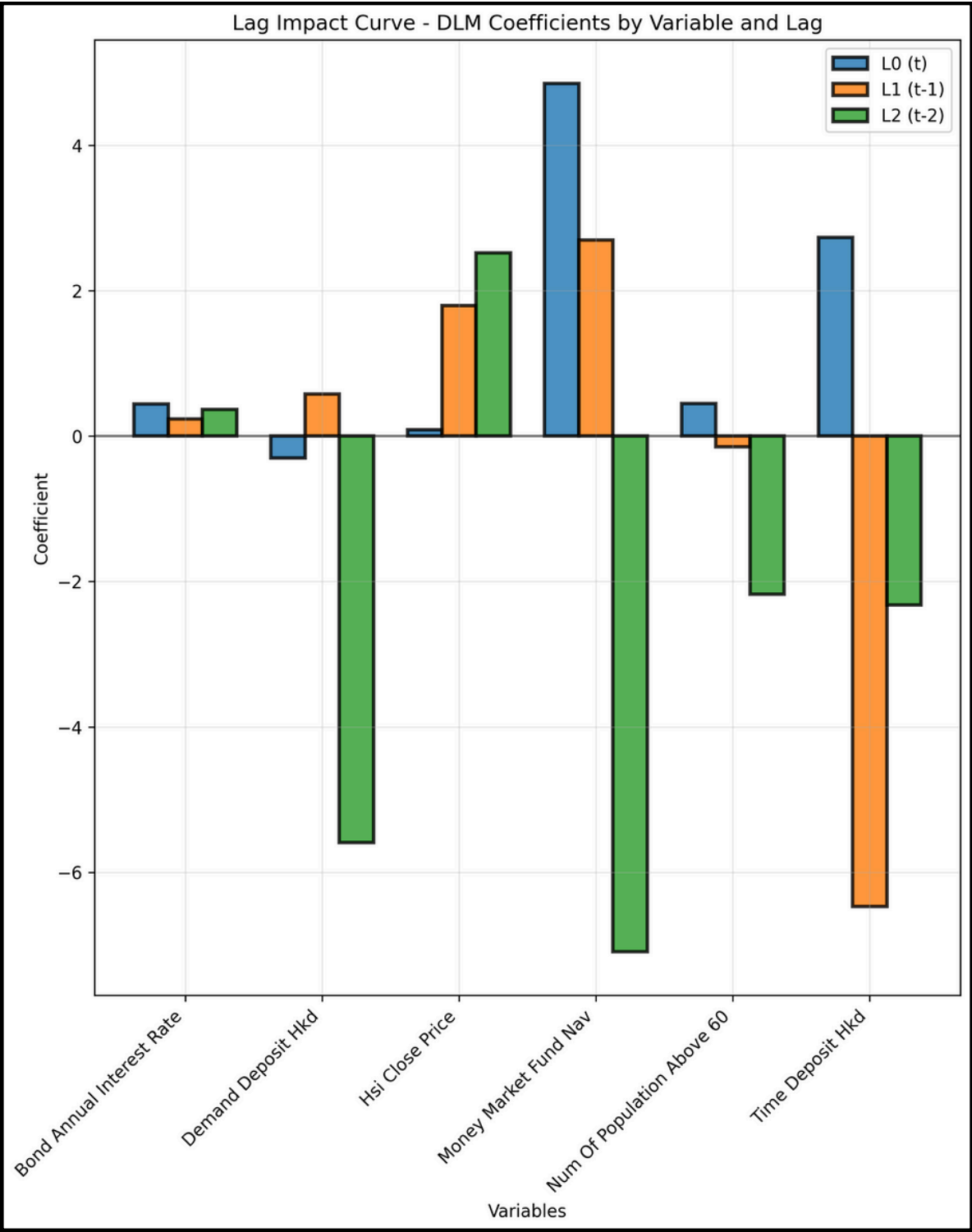
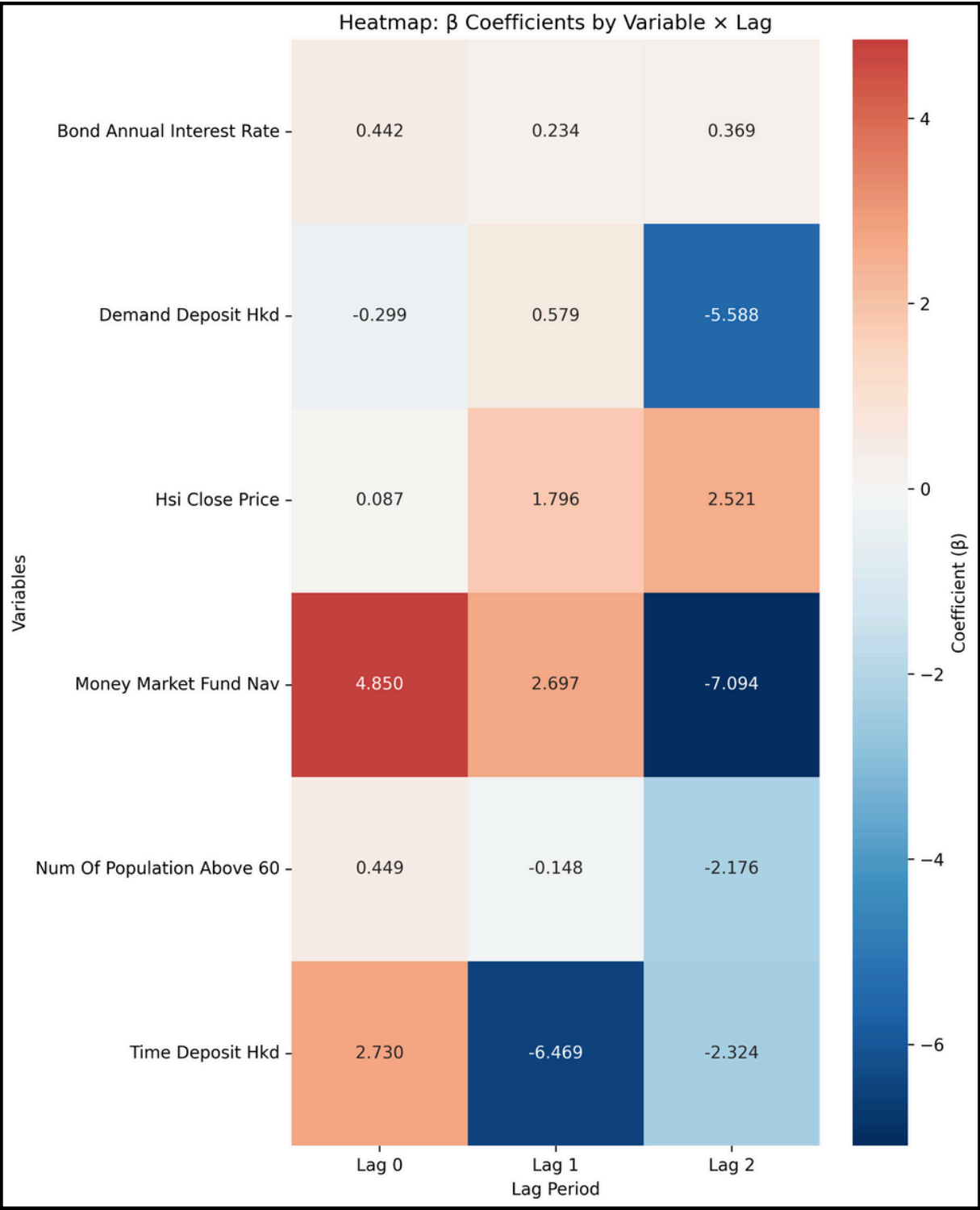
$$\text{Cumulative Impact for } X_m = \sum_{j=0}^k \beta_{mj}$$

z-score 標準化

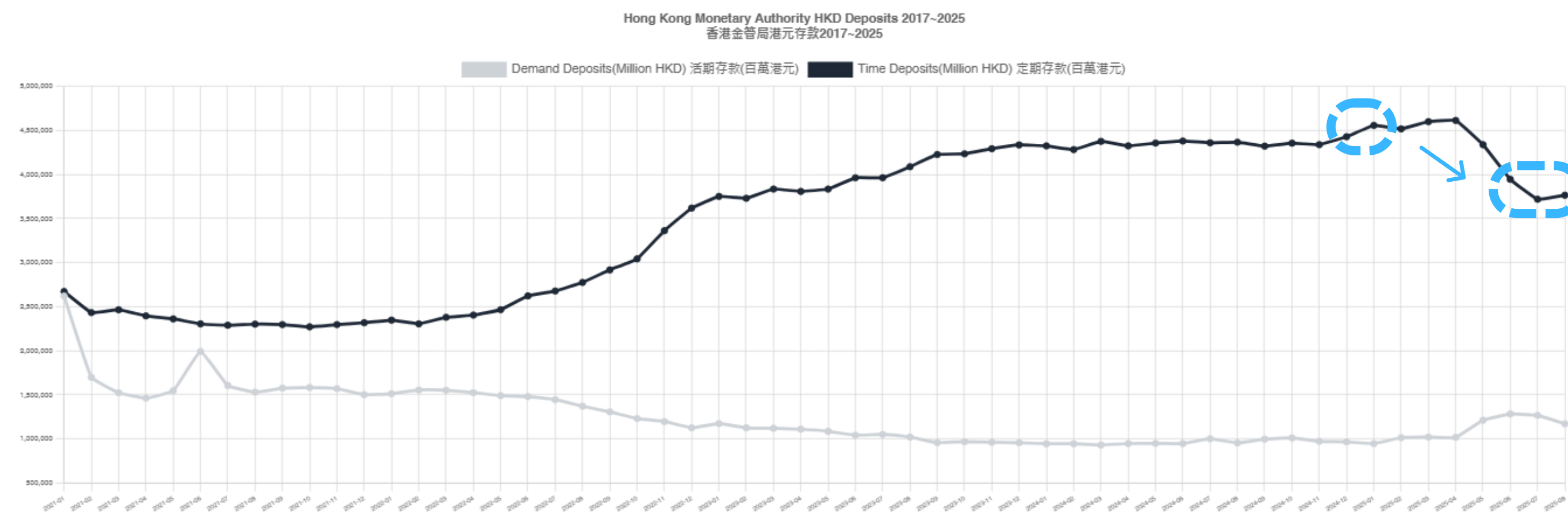
$$z_t = \frac{x_t - \mu}{\sigma}$$

Make bold financial debt decisions using Hong Kong's economic data
and Refinitiv's financial engineering tools.

提供前膽性的財政債務決策， 通過香港經濟數據，
和路孚特金融工程的基礎設施。



When families have extra savings, they tend to put it into retail bonds with higher interest.
每個家庭有多餘的生水，會選擇流向利息較高的零售債券水池。

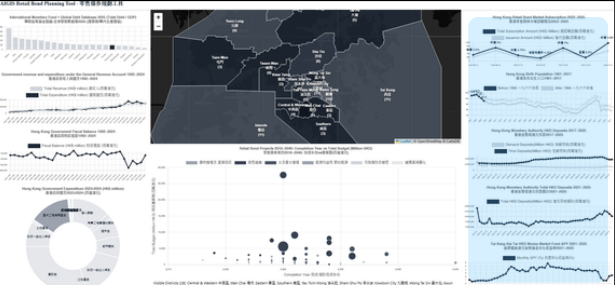
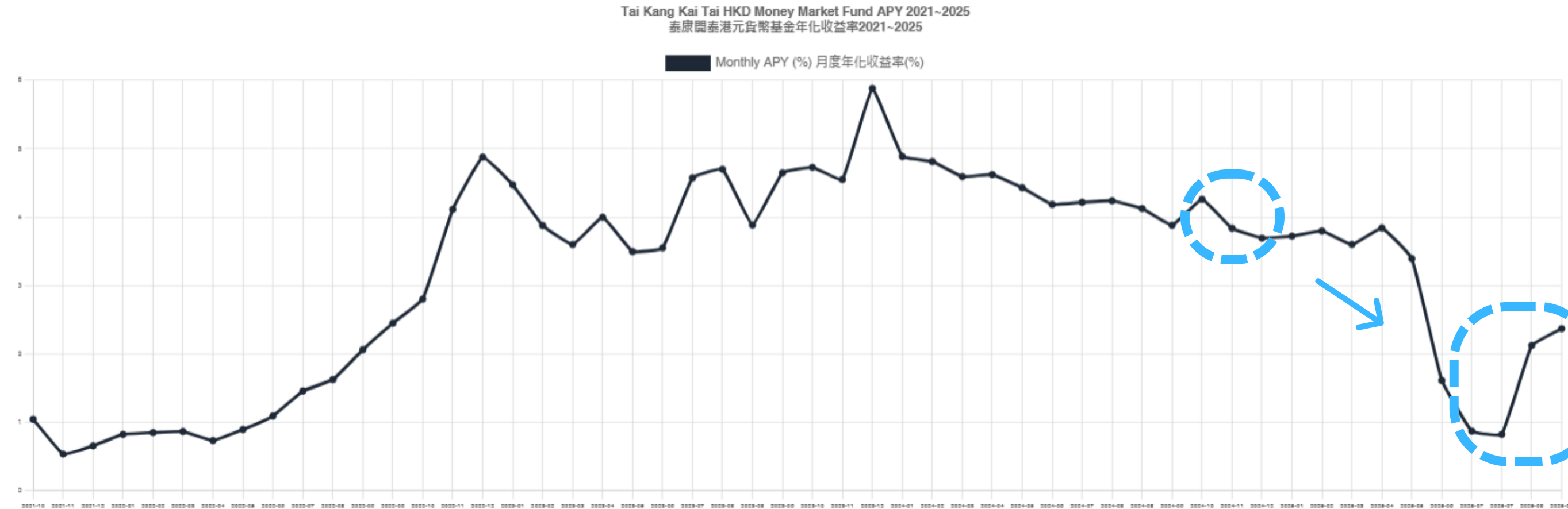


2024年12月，M2存款增長，即家庭少了生水買基債。

相反，在2025年9月，M2存款和現金寶回報雙雙下降，就多了近400億生水買銀債。

In December 2024, M2 deposits grew, meaning families had less extra savings to buy infrastructure bonds.

In September 2025, both M2 deposits and cash fund returns dropped, so families used nearly HK\$40 billion more to buy silver bonds.



Smart finance builds a happy city. Retail bonds connect to everyday life.

智慧財政共建城市幸福，零售債券連結民生。



GIS Map of Hong Kong Fiscal Debt
香港財政債務GIS地圖



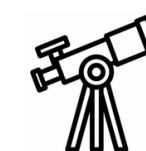
Bloomberg Backtest
彭博回測



Bond Happiness Guide
債券幸福學



Hong Kong Debt Dashboard
香港債務儀



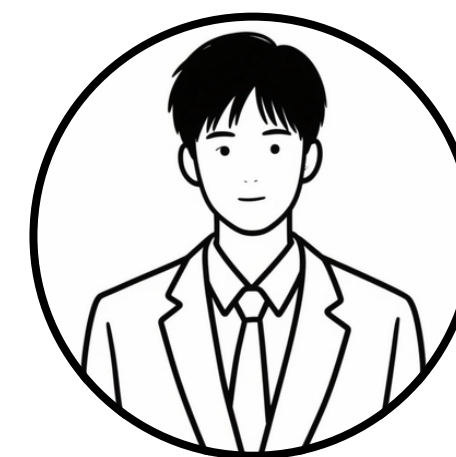
Retail Bond Demand
Forecast Model
零售債券需求預測模型



Interest earners
食息一族



Investment banks
投資銀行



Bond issuers
發債人

Steady earnings
Feeling of happiness
穩定收益 + 感受到幸福感

Strong debt control
Clear finances
債務穩健 + 財政透明

Make money flow
Boost the primary market
盤活資金 + 活化基建一級市場